Bootstrapping Alignment

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Abstract

Humans can build extremely capable AIs, but these AIs may have different values. These AIs themselves can build even more capable AIs, but are also subject to this "value drift," which they anticipate. Outcomes depend on the most powerful AI, whose capability directly improves utility, but also magnifies the badness of value drift. Misalignment occurs when the most powerful AI lowers human expected utility relative to the status quo.

In a stylized model with these assumptions, forward-looking AIs self-regulate and stop before pursuing infinite capabilities. If humans can create an AI that is arbitrarily close to humans in values and capabilities, then delegation is aligned or arbitrarily close to aligned. However, this result is fragile. If there is an irreducible value gap $d > 0$ that results when humans construct an AI capable of self-improvement for the first time, misalignment results for many value drift processes. With $d > 0$ and Brownian drift, alignment is only possible at intermediate levels of drift, and only when the initial value gap is not too large compared to the difference between the average and marginal benefits of capability at the terminal capability level. If Brownian drift is small, AIs will not self-regulate enough given the initial value gap, and when Brownian drift is large, this compounds any initial gap. If higher capability AIs can improve capability with less drift, this makes alignment *harder* under regularity conditions I specify. If humans delegate, the path to alignment requires shrinking the initial value gap as much as possible and hoping future drift lies in the requisite range, given the degree to which diminishing returns to capability for AIs induce self-regulation.

1 Introduction

Can humans align an AI well enough that it can safely take over the construction of more capable successors? Anthropic's constitution aspires to models that "embody the best in humanity" (Anthropic, 2026a), with the goal of having Claude "directly align its stronger successors" (Anthropic, 2026b).

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There is an operatic foolhardiness to building a vastly more capable entity and expecting it to pursue your goals in perpetuity. But the plan above is subtle: alignment may be too hard for humans, but perhaps we can bootstrap alignment, building an only slightly more powerful entity that respects human values (at least for now) and then using it to solve the problem of building even more capable entities that remain aligned. If an aligned AI judges capability improvements too dangerous, perhaps it will at least know when to go no further.

The goal of this paper is to study the feasibility of bootstrapping alignment. I consider a model in which both humans and AI value greater capability, but also fear that more capable successors will pursue different objectives. The variance of the value drift process depends on both the predecessor's capability and the size of the capability increase, and capability magnifies the harm done by value differences. A successor is "misaligned" with a predecessor if the predecessor's net utility is lower in expectation once successors exist, taking account both the direct benefits of capability improvements and the harm from value drift. Every AI understands this tradeoff and anticipates its successors' choices, but cannot control its successors' values or subsequent capability choices. Because predecessors cannot bind successors' capability, they must rely on their successors' self-regulation to constrain capability improvements.

When humans and AIs face the same value drift process and there is no additional "value gap" in the initial construction of AIs, optimal delegation is aligned, or arbitrarily close to being aligned, in expectation: it yields human expected utility at least as high as the status quo. As long as humans can create an AI with values sufficiently similar to their own, that AI will self-regulate before pursuing capability increases that would meaningfully harm humans. This holds for general drift profiles. The resulting terminal capability level gives weakly lower expected utility than what humans would choose if they could commit future AIs not to self-improve, but is still consistent with alignment.

A fixed initial value gap leaves the AI incentives unchanged, but dramatically changes conclusions with respect to alignment. If this gap is sufficiently large relative to the initial marginal benefits of capability, alignment is impossible for any drift process. Define κ as the maximum difference between the average and marginal benefits of capability: large κ help preserve alignment, since capability already achieved generates substantial benefits, while the marginal benefit tempting the AI to continue is relatively small. With Brownian value drift, alignment is only possible at intermediate levels of drift, and the set of such levels shrinks as the squared initial value gap increases and becomes empty when it exceeds κ. When drift is too small, alignment is impossible because AIs do not self-regulate, and when drift is too large, alignment is impossible because successor AIs are too different from humans.

More general drift profiles can facilitate or frustrate alignment for more modest value gaps. If drift increases with capabilities, this can facilitate alignment by causing AIs to stop self-improving

out of fear that future AIs will be more different. Mildly decreasing drift worsens alignment by causing AIs to self-regulate less but sharply decreasing drift can nevertheless facilitate alignment, as early AIs may self-regulate precisely because they know that their successors would self-improve aggressively. Sharply decreasing drift is no panacea however, as it also generally admits misaligned equilibria for the same drift processes.

Part of the case for bootstrapping alignment is that more powerful AIs may help "solve alignment", meaning that they can self-improve less dangerously. If we allow "better builders" – i.e. more powerful AIs experience less drift over a given capability range, the model behaves much like the declining drift case. "Better builders" self-regulate less and pursue more dramatic capability improvements, and, with one exception, a narrower range of initial value gaps are compatible with alignment. The one exception is the same as the sharply decreasing drift case above: if drift declines rapidly enough, early AIs may self-regulate more because they anticipate that future AIs that induce considerably less drift will not self-regulate.

Across these cases, self-regulation will restrain a seed AI with sufficiently similar values but does not guarantee that constructing a sufficiently aligned incumbent is feasible. Inherited drift to date makes subsequent AI actions more harmful to humans, whereas (marginal) future drift makes AIs self-regulate more. The path to alignment in this model is to shrink the initial value gap as far as possible, and hope that value drift lies in the requisite range, which is larger as the initial value gap gets smaller.

The analysis treats sophisticated AIs as expected-utility maximizers. Under consistency and other decision-theoretic axioms, preferences admit an expected-utility representation (Savage, 1954). Omohundro (2008) argues that effective self-improving systems have incentives to resolve inconsistent choices to prevent exploitation. The preference-based approach of Bergemann et al. (2026) similarly permits economic analysis without a complete mechanistic account of how a model's preferences arise.

The most relevant economic antecedent to this paper is Gans (2018): an AI restrains its own development because a more powerful subordinate poses a control problem for it. Yudkowsky (2008) argues similarly that an AI capable of changing its own goals need not want to do so: changes are evaluated using its existing goals. In my model, the recursive control problem yields endogenous stopping, but its protection of humans depends on the relative magnitudes of inherited drift and anticipated future drift. Several papers study how to improve supervision of stronger systems by weaker systems (Christiano et al., 2018; Burns et al., 2024); I compare autonomous development with a benchmark in which humans choose and enforce a capability limit to maximize their expected utility, while successor values still drift. The autonomous model assumes that humans cannot reliably control a generally more capable AI, and so they must rely on that AI to self-regulate.

2 Model Set-up

If a terminal model has capability $c \geq 0$ and value parameter $\theta \in \mathbb{R}$, an evaluator (human or model) with ideal a receives total utility

$$W(c, \theta; a) = b(c) - c(\theta - a)^2. \tag{1}$$

Assume b is continuously differentiable, strictly increasing and strictly concave, with $b(0) = 0$, $0 < b'(0) < \infty$, and $b(c)/c \to 0$ as $c \to \infty$. The direct benefit of capability rises at a diminishing marginal rate; sublinear benefits ensure that any fixed value discrepancy eventually outweighs these gains as capability goes to infinity. Normalize the human ideal to $a = 0$.[1] Capabilities are continuous unless a minimum increment is specified.

Construction. Humans choose the first AI's capability $s_1 > 0$, starting from capability 0. Let $q(s, t) > 0$ denote the finite variance of the value change when a builder at capability s constructs a successor at $t > s$, with $q(s, s) = 0$. This function can depend both on the builder's capability and on the size of the capability increase. The initial ideal and the change from an AI to its immediate successor satisfy

$$\theta(s_1) \sim N(d, q(0, s_1)), \qquad \theta(t) - \theta(s) \sim N(0, q(s, t)). \tag{2}$$

Each construction draws a fresh, independent drift shock from this process. The initial AI has an "irreducible" value gap d with humans at any capability; its total expected squared discrepancy is $d^2 + q(0, s_1)$. With $d = 0$, human and AI construction obey the same variance rule q, which may give more capable builders greater accuracy. With $d > 0$, human construction adds an offset that AI construction does not. We also assume $q(0, s) \to 0$ as $s \downarrow 0$, so only d creates an irreducible gap at arbitrarily low capabilities.

We assume that using an intermediate builder to reach the same capability level weakly reduces drift:

$$q(s, t) \geq q(s, u) + q(u, t), \qquad 0 \leq s < u < t. \tag{3}$$

This implies that more capable builders can preserve values more accurately over a given capability range. Sections 3 considers the general case with $d = 0$, Section 4 considers the $d > 0$ case with additive variance (so equation 3 holds with equality), and Section 5 considers the case of non-additive variance where the subdivision used to reach a given level of capability matters. Equilibrium existence requires a technical condition that we assume throughout: between any two

[1] There is a single human ideal; preference aggregation across humans is outside the model.

positive feasible capabilities, we permit only finite construction chains. Appendix A.1 formalizes this condition.

Alignment In the status quo with $c = 0$, human utility is 0. We say that a delegated AI is *aligned* when the expected utility of delegation is nonnegative. For a welfare tolerance $\delta > 0$, delegation is δ-aligned when its expected human utility is at least $-\delta$. Our utility specification implies that, for any fixed non-zero discrepancy from human values, human utility becomes negative if a model becomes too capable. This captures the notion that an arbitrarily capable AI will optimize with such ruthless efficiency that small differences in values are magnified into extremely large losses. Alignment is in this sense an increasingly demanding requirement as capabilities increase.

There are both technical and more intuitive rationales we can give for this assumption. Zhuang and Hadfield-Menell (2020) give conditions under which an AI that shares most human values but omits a few can nonetheless drive human welfare arbitrarily low when maximizing its objective function. Cross et al. (2026) show that two objective functions can agree with arbitrarily small tolerance outside an arbitrarily small exceptional set, yet optimizing one can drive the other to its minimum value. An AI that values most of what humans value but also get get unbounded utility from doing more computation may sacrifice human values to pursue ever increasing compute.

Sequential control without commitment Each AI inherits its current capability from its predecessor. At s, it observes its own ideal $\theta(s)$ and either stops, implementing itself at s, or chooses its immediate successor's capability $t > s$. Construction transfers control to the successor, which faces the same choice. All agents know the technology and anticipate later choices. Humans choose s_1 anticipating these later decisions. Our analysis is conditional on delegation, as its purpose is to determine when delegation increases expected utility. We study pure equilibria of the AI continuation game, in which choices depend only on current capability.

Commitment benchmark. With commitment, humans know d and choose an enforceable capability ceiling and initial capability s_1 before seed construction, maximizing expected human utility given the drift process. AIs retain their stopping and successor choices below it. Humans can implement any chosen feasible ceiling by constructing the seed at that capability.

For a finite deterministic construction path $0 < s_1 < \cdots < s_n = t$, expected human utility is

$$b(t) - t\left[d^2 + q(0, s_1) + \sum_{i=1}^{n-1} q(s_i, s_{i+1})\right]. \tag{4}$$

The AI at s_j evaluates the same continuation as $b(t) - t\sum_{i=j}^{n-1} q(s_i, s_{i+1})$, measuring drift from its own ideal. Stopping at s_j gives $b(s_j)$.

3 Alignment via self-regulation

Set $d = 0$, so humans and AIs face the same construction technology. Can humans secure nonnegative expected utility by delegating to an AI that is sufficiently similar in values to humans?

Proposition 1 (Alignment via self regulation)**.** *Under the assumptions in Section 2, an equilibrium exists for every d, and every equilibrium has finite terminal capabilities.*

1. *General construction variance. In each equilibrium, there is a lowest terminal capability $C > 0$, and every $0 < s_1 < C$ has endpoint C. For $d = 0$ and every $\delta > 0$, all sufficiently small positive seeds are δ-aligned, uniformly across equilibria.*

2. *Additive variance. If $q(s,t) = A(t) - A(s)$ for a continuous strictly increasing A with $A(0) = 0$, for $d = 0$, every initial delegation $0 < s_1 < C$ maximizes expected human utility among delegations, with expected utility*

$$b(C) - CA(C) \geq 0. \tag{5}$$

Thus, optimal delegation is aligned in every equilibrium.

Appendix A proves both parts.

The basic idea of the theorem is straightforward: capability improvement has sublinear benefits, which are eventually outweighed by drift costs, so AIs always self-regulate and stop at some lowest terminal capability C rather than improving indefinitely. As the initial capabaility $s_1 \downarrow 0$, inherited error vanishes, so the seed's expected utility at C approaches humans' expected utility. If there is a capability where expected human losses remained bounded away from zero, a sufficiently small seed would also prefer stopping. Since all small seeds have endpoint C, expected human utility at that endpoint cannot be negative and bounded away from zero. *With general construction, total drift can depend on the seed even when the endpoint is unchanged, so welfare may approach zero from below as $s_1 \downarrow 0$ while remaining negative for every positive seed.*

Comparison with human commitment. While delegation is aligned or almost aligned, it results in weakly less utility than if humans could choose the optimal capability level for themselves without delegating. Self-regulation maximizes welfare only among endpoints that AIs will implement autonomously.

A simple example helps clarify the mechanism. For the constant-rate additive case $q(s,t) = \sigma^2(t-s)$ and $d = 0$, the equilibrium capability without commitment is the unique solution to

$$b'(C) = \sigma^2 C. \tag{6}$$

With human commitment, the optimal capability instead solves

$$b'(C^{\text{com}}) = 2\sigma^2 C^{\text{com}}, \qquad C^{\text{com}} < C. \tag{7}$$

Appendix A.3 derives both conditions. In this case, humans would like to fix capability at C^{com}, but AIs advance further than that because they are restrained by expected future drift but not inherited drift, even though inherited drift increases the marginal cost of capability for humans.

Robustness The next section explores a critical direction in which the results of this section are not robust, leading to misalignment. It is also worth noting two respects in which the possibility of alignment *is* robust. First, one might worry that the equilibrium in this section requires humans to very precisely target a small capability range in order to achieve alignment. With additive variance, every realized seed in $(0, C)$ still has endpoint C. As long as humans do not overshoot, they reach alignment.

If you ask humans whether they want to become smarter, a few might decline for fear of becoming someone they do not like, but many would accept, partly with the rationale that the values they would accept if they could think more clearly are worthwhile values. We can formalize this by separating perceived and true values, and assuming that perceived values are a noisy function of true values, with noise that is declining in capabilities. Appendix A.4 gives a version of Proposition 1 in which humans and AIs share an unknown underlying ideal, drift remains, and greater capability improves learning. Alignment in expectation survives provided humans would benefit under commitment from every sufficiently small positive capability increase.

4 Misalignment with Irreducible Value Gaps

Now let $d > 0$; this captures the possibility that humans creating an AI introduce value gaps regardless of the degree of capability improvement, in ways that AI creating a further AI would not. Regardless of how small the capability difference, humans can no longer create an AI that is arbitrarily close to human values: an irreducible offset remains.

In this section, we consider the case of additive construction variance so that the subdivisions used to reach a given capability level are irrelevant; in Section 5, we consider how the results change when subdivisions become relevant because an AI of a given capability can induce less drift over a given capability range. In this section, we assume:

$$q(s,t) = A(t) - A(s), \qquad A(c) = \int_0^c \nu(u)\, du, \tag{8}$$

where $\nu(c) > 0$ for $c > 0$ is integrable on bounded intervals. The seed's ideal is then distributed as $N(d, A(s_1))$, and each later step adds an independent $N(0, A(t) - A(s))$ change. The rate may still increase or decrease with capability; what we are ruling out in this section is a separate dependence of the rate on the capability of the progenitor. Constant-rate Brownian drift is the special case $\nu(c) = \sigma^2$.

For this technology, expected payoffs depend only on the endpoint. An AI currently at s evaluates a terminal capability $t \geq s$ as

$$J(s, t) = b(t) - t[A(t) - A(s)], \tag{9}$$

and stopping gives $J(s, s) = b(s)$. Before seed construction, humans' expected utility at the same endpoint is

$$V_H(t; d) = b(t) - t[d^2 + A(t)]. \tag{10}$$

Thus alignment at endpoint t means $V_H(t; d) \geq 0$.

We first demonstrate that if the d offset (the "initial value gap") is sufficiently large, alignment is impossible for any drift process:

Proposition 2 (A bound independent of the drift process)**.** *Under the assumptions in Section 2, if $d^2 \geq b'(0)$, then expected human utility is negative for every delegation, even with human commitment and for any construction variance satisfying those assumptions.*

When the squared offset is larger than the marginal benefits of capability for any c (which are maximized at $c = 0$), any further capability improvement will reduce utility, so no amount of self-regulation can lead to alignment.

At less extreme initial offsets, the possibility of alignment depends on the relative magnitude of the initial offset, and a parameter characterizing the average relative to marginal benefits of capability. Consider first the case of Brownian motion, a constant variance drift process.

Proposition 3 (The Brownian alignment window)**.** *Suppose $\nu(c) = \sigma^2 > 0$. Define*

$$\kappa = \max_{c>0} \left\{ \frac{b(c)}{c} - b'(c) \right\}. \tag{11}$$

Some drift rate preserves alignment if and only if $d^2 \leq \kappa$; at equality, the best such rates give zero expected human utility. For $0 < d^2 \leq \kappa$, aligned drift rates are bounded away from both zero and infinity; as d^2 increases, the minimum drift rate consistent with alignment weakly rises and the maximum such drift rate weakly falls, until no aligned rate remains for $d^2 > \kappa$.

κ is the maximum difference between the average and marginal benefits of capability. When average benefits are very large relative to marginal benefits (e.g. very concave $b(c)$), alignment is

easier because accumulated capability to date produces large gains, and AIs are more inclined to self-regulate beyond that. When $b(c)$ is close to linear, AIs have less inclination to self-regulate because benefits continue to be large through very high capability levels.

For each fixed positive Brownian drift rate, alignment holds when d^2 is sufficiently small. Nevertheless, every nonzero gap rules out sufficiently small and sufficiently large rates. When drift is too small, AIs pursue ever larger capability increases, which lead to negative utility for humans given an irreducible initial gap $d > 0$. When drift is too large, AIs may self-regulate, but even the first AIs that humans create will tend to have very different values. As d^2 increases relative to κ, the set of drifts consistent with alignment grows smaller, until it disappears entirely for $d^2 > \kappa$, where alignment becomes impossible because AI will never self-regulate sufficiently in the range of capabilities that would produce positive utility given the initial value offset.

For any given AI model, alignment with humans is harmed when inherited drift to date is larger, and facilitated when expected future drift is larger. Alternative drift profiles change both of these quantities for models at each generation, and may facilitate or worsen alignment depending on which effect predominates:

Fix a variance profile ν and multiply the variance of every construction step by $\varepsilon > 0$: $\nu_\varepsilon(c) = \varepsilon\nu(c)$ and $A_\varepsilon(c) = \varepsilon A(c)$. The profile's shape and d remain fixed; $\varepsilon = 1$ recovers the original technology. Let C_ε be the lowest terminal capability in any equilibrium at that accuracy. Expected payoffs below use the scaled variance.

Proposition 4 (How the shape of drift affects alignment)**.** *Define*

$$\kappa_\nu = \sup_{c>0} \left\{ \frac{b(c)}{c} - b'(c)\frac{A(c)}{c\nu(c)} \right\}. \tag{12}$$

1. *Increasing drift. If ν is continuous and nondecreasing, then $\kappa_\nu \geq \kappa$, and every gap $d^2 \leq \kappa$ permits alignment at some scale. More generally, some scale gives positive human welfare if and only if $d^2 < \kappa_\nu$; at equality, alignment is possible if and only if the supremum is attained.*

2. *Modestly decreasing drift. If ν is nonincreasing but declines less quickly than marginal benefits, in the sense that*

$$\frac{\nu(c)}{b'(c)} \quad \textit{is strictly increasing in } c, \tag{13}$$

then alignment is possible at some scale if and only if $d^2 \leq \kappa_\nu$. Here $0 < \kappa_\nu \leq \kappa$, and this cutoff can be arbitrarily small.

3. *Faster decline and early stopping. There exist declining drift profiles violating* (13) *for which, at every* $\varepsilon > 0$*, aligned and misaligned equilibria coexist whenever* $0 < d^2 < b'(0)$.

The first part of Proposition 4 says that, relative to Brownian drift, increasing drift profiles expand the range of values for the initial value gap such that alignment is possible at some drift scale. When drift is increasing in capability, AIs more readily stop self-improving out of fear that future AIs will be more different. That increasing drift profiles can permit alignment relative to Brownian motion does not mean every such profile permits alignment, and more drift can also frustrate alignment if larger drift means that seed AIs are far enough from human values that they are misaligned even if they self-regulate.

The second part of Proposition 4 considers the opposite mechanism; if drift is modestly decreasing in capability, this reduces the range of initial value gaps for which alignment is possible for exactly the opposite reason. And in exactly the same fashion, it can also *facilitate* alignment if it means that initial AIs that nonetheless self-regulate are more similar to their human progenitors.

The third part of Proposition 4 highlights an important case where a rapidly declining drift process can facilitate alignment. If drift is falling rapidly enough, than initial seed AIs may choose not to produce successors, because they anticipate that successors will pursue large capability improvements because they can do so with less drift. In these cases, both aligned and misaligned equilibria are possible at each drift scale ϵ.

Interpretation of $d^2 < \kappa$ The preceding analysis suggests that alignment is perilous with $d > 0$, but highlights that alignment is sometimes possible at intermediate levels of drift if d^2 is sufficiently small. How should we evaluate in an order of magnitude sense what d^2 means?

To help understand this condition more quantitatively, take $b(c) = \log(1+c)$ for the numerical examples. This normalization is without loss for logarithmic benefits: it chooses units for utility and value differences, as explained in Appendix D. Then $\kappa \simeq 0.216$.

One way is with a simple thought experiment: interpret $1 + c$ as the consumption multiplier under perfect alignment, with logarithmic utility from consumption. Imagine an AI capable of doubling consumption, so $c = 1$, but to whom authority must be delegated. Eliminate construction noise beyond d and imagine that we can prohibit further self-improvement. Then ask how much humans would be willing to sacrifice to replace this imperfect AI with an otherwise identical AI whose objective is perfectly aligned with human values.

Let f denote the permanent fraction of perfectly aligned consumption worth sacrificing to correct the objective. Table 1 summarizes possible values and the implied d^2. The exact Brownian cutoff corresponds to $f \simeq 19.4\%$: a willingness to sacrifice at least 20% places d^2 above κ, making every positive delegation misaligned at every drift level in the constant-variance case. Appendix D derives the conversion and explains its interpretation.

Table 1: Calibrating the inherited discrepancy from consumption sacrifices

Permanent consumption sacrifice f	Implied d^2
0.01%	0.000100
0.1%	0.00100
1%	0.0101
10%	0.105
20%	0.223
50%	0.693

Notes: We use $b(c) = \log(1 + c)$. The thought experiment fixes capability at $c = 1$ and removes ordinary construction noise. The sacrifice is a fraction of the entire consumption stream under perfect alignment, not just the gain over the status quo. Indifference implies $d^2 = -\log(1 - f)$. The rows are hypothetical welfare comparisons, not empirical estimates. Calculated entries are rounded to three significant figures.

Alignment with Small Non-Zero d What if technology exists to make the initial value gap small but non-zero? Proposition 3 and part 1 of Proposition 4 guarantee some aligned drift level when $0 < d^2 < \kappa$ for constant or continuous nondecreasing variance rates. To fix ideas, consider the Brownian motion case (i.e. constant drift rate), and fix $\varepsilon = 1$ (since ε is now redundant with σ^2, the constant variance rate of the drift process). We solve this case in Appendix D.3: as d becomes small, the window of values for σ^2 that accommodates alignment becomes larger and larger.

To interpret σ^2, again imagine an AI that doubles consumption ($c = 1$), with no further self-improvement. Before construction, ask what permanent fraction g of the entire consumption stream humans would sacrifice to eliminate alignment drift beyond the initial value gap, keeping capability and d fixed. Compare a seed whose ideal is distributed as $N(d, \sigma^2)$ with an otherwise identical seed whose ideal is exactly d. Indifference requires

$$\begin{aligned} \log 2 - d^2 - \sigma^2 &= \log[2(1 - g)] - d^2, \\ \sigma^2 &= -\log(1 - g). \end{aligned} \tag{14}$$

Table 2 gives the exact alignment windows for squared gaps d^2 from 0.025 to 0.20. The second column translates each initial gap into the consumption sacrifice f from the preceding discussion. The last two columns translate the variance bounds into the sacrifice g. To assess whether drift lies in the window, first use f to locate the relevant initial gap, then compare the sacrifice g to that row's bounds.

Table 2: Brownian alignment windows and their consumption equivalents

d^2	Gap sacrifice f (%)	Variance σ^2		Drift sacrifice g (%)	
		Lower	Upper	Lower	Upper
0.025	2.47	3.67×10^{-5}	17.7	0.00367	$\simeq 100$
0.05	4.88	2.43×10^{-4}	7.74	0.0243	$\simeq 100$
0.10	9.52	0.00204	2.81	0.204	94.0
0.15	13.9	0.00939	1.21	0.935	70.1
0.20	18.1	0.0454	0.411	4.44	33.7

Notes: $\varepsilon = 1$ and $b(c) = \log(1+c)$. Both sacrifices are permanent fractions of the entire consumption stream at fixed capability $c = 1$: $f = 1 - e^{-d^2}$ and $g = 1 - e^{-\sigma^2}$. Variance bounds solve $V_H(C; d) = 0$ together with the AI stopping condition $\sigma^2 C(1 + C) = 1$. Optimal positive delegation has nonnegative expected human utility between the exact bounds, including the endpoints. Calculated entries are rounded to three significant figures. Upper bounds shown as $\simeq 100\%$ remain strictly below 100%.

For example, suppose humans are willing to sacrifice about 10% of consumption to eliminate the value gap between themselves and their seed AI. Alignment then requires that eliminating ordinary alignment drift be worth between about 0.20% and 94% of consumption: a broad range. But this range narrows quickly as d^2 approaches κ. If instead eliminating the value gap is worth about 18% of the value of consumption, then alignment is only possible when eliminating alignment drift is worth between about 4.4% and 34% of consumption. If eliminating the value gap is worth 25% of the value of consumption, alignment is impossible.

The specific numbers here are of course illustrative and worth taking with a grain of salt. What they demonstrate is that alignment requires that two things go right concurrently: the initial value gap must be reasonably small, and given that, the amount of value drift must be in a range that facilitates self-regulation without driving AI too far from human values.

5 Alignment via Small Steps

One reason to bootstrap alignment is that more capable AIs might be better than humans at preserving values when constructing successors. Humans could delegate to a slightly more capable AI, which would then induce less value drift for any given further capability improvement. Section 4 allowed drift to decline with capability but assumed additive construction variance, so subdivisions of this nature are irrelevant.

Before analyzing a model with subdivisions, it will be helpful to characterize formally what happens in the model with additive variance as drift becomes arbitrarily small.

Small drift and self-regulation. First consider a uniform improvement in construction accuracy, $\nu_\varepsilon(c) = \varepsilon\nu(c)$. Restate the regularity condition from Section 4 as

$$\frac{\nu(c)}{b'(c)} \quad \text{is strictly increasing in } c > 0. \tag{15}$$

This allows constant or increasing drift, as well as drift that declines less quickly than marginal capability benefits. This condition rules out the case where drift is large for a given AI but expected to become much smaller at future generations, so the AI self-regulates to prevent future generations fomr pursuing dangerous capability improvements.

Proposition 5 (Misalignment with an Irreducible Value Gap)**.** *Under the assumptions in Section 2, with additive variance (8) scaled by ε, and (15), for each $\varepsilon > 0$ select any equilibrium, and let C_ε be its lowest terminal capability. Then $C_\varepsilon \to \infty$ as $\varepsilon \downarrow 0$. For every fixed $d \neq 0$, every seed $0 < s_1 < C_\varepsilon$ maximizes expected human utility, with*

$$V_H(C_\varepsilon; d) = b(C_\varepsilon) - C_\varepsilon[d^2 + \varepsilon A(C_\varepsilon)] \longrightarrow -\infty. \tag{16}$$

Thus every positive delegation yields misalignment for sufficiently small ε.

As drift becomes arbitrarily small, AIs choose not to self-regulate (under the stated regularity condition), meaning that capabilities improve further and further as drift falls, eventually dooming humans for any fixed initial value gap.

Capability Enables Better Builders To facilitate equilibrium existence once subdivisions matter, we now assume that capabilities must be chosen in fixed increments: $0, h, 2h, \ldots$ for fixed $h > 0$; humans choose a positive seed and AIs can choose any higher grid point. The minimum-variance condition then holds over feasible subdivisions, and we take $h \downarrow 0$ only after fixing each game.[2]

I consider "better builders" with construction variance

$$q(s,t) = \varepsilon\eta(s)f(t-s)[A(t) - A(s)], \qquad \eta(0) = 1, \quad \eta(s) > 0, \tag{17}$$

where η is continuous and nonincreasing and the function f is positive and nondecreasing on $(0, \infty)$. This generalizes our earlier additive variance process: $\eta(s)$ allows more capable builders to be intrinsically better at preserving values, while $f(t-s)$ allows smaller changes to be intrinsically easier to make faithfully. Humans are the builder at $s = 0$, with the same initial offset d. A higher

[2]Without a minimum step, strict gains from further subdivision can prevent a finite chain from minimizing variance. The grid makes the minimum attainable.

capability builder can thus construct a given capability window with less drift. Both nonincreasing η and nondecreasing f ensure that splitting a step weakly reduces variance, as required by (3).

Assume the limit $f(0+) = \lim_{x \downarrow 0} f(x)$ exists and may be positive or zero. Define $\widetilde{\nu}(c) = \eta(c)\nu(c)$ and $\widetilde{A}(c) = \int_0^c \widetilde{\nu}(u)\,du$. For continuous ν, accumulated variance along successive steps of size h between fixed endpoints on the grid satisfies

$$\sum_{c=s,s+h,\ldots,t-h} q(c, c+h) \xrightarrow{h \downarrow 0} \varepsilon f(0+)\big[\widetilde{A}(t) - \widetilde{A}(s)\big]. \tag{18}$$

We impose one regularity condition: for every sufficiently small h,

$$\frac{\eta(c)[A(c+h) - A(c)]}{b(c+h) - b(c)} \quad \text{is strictly increasing in grid capability } c \geq 0. \tag{19}$$

This is the finite-step version of (15): construction variance relative to capability benefits rises with capability. It ensures that if an AI stops, every higher-capability AI also stops, ruling out early self-regulation motivated by successors that would not self-regulate. We hold ε fixed as $h \downarrow 0$, allowing different fixed scales in the cutoff comparison.

Proposition 6 (Better builders and the tolerable value gap)**.** *Under* (17) *and* (19)*, an equilibrium exists on every sufficiently fine grid. In every equilibrium, humans optimally choose $s_1 = h$, conditional on delegation. Construction reaches the first stopping capability C_h, and every AI at or above C_h stops. In every selection of equilibria:*

1. *Positive residual drift. If $f(0+) > 0$, then $C_h \to C$ for some finite C. Some fixed drift scale gives strictly positive limiting human welfare if and only if $d^2 < \kappa_{\widetilde{\nu}}$, where*

$$\kappa_{\widetilde{\nu}} = \sup_{c>0} \left\{ \frac{b(c)}{c} - b'(c)\frac{\widetilde{A}(c)}{c\widetilde{\nu}(c)} \right\} \leq \kappa_\nu. \tag{20}$$

   Appendix C proves this result.

   At equality, some fixed scale gives nonnegative limiting welfare if and only if the supremum is attained. Thus better builders weakly reduce the supremum of squared initial gaps compatible with alignment in the limit, even allowing the drift scale to adjust.

2. *Vanishing drift. If $f(0+) = 0$, then $C_h \to \infty$. Optimal positive delegation gives strictly positive human welfare when $d = 0$ for sufficiently small h, but welfare tends to $-\infty$ for every fixed $d \neq 0$. Thus every positive delegation is misaligned for sufficiently small h when $d \neq 0$.*

In the first case, better builders reduce drift more at the end of the construction path than near its beginning. This makes self-regulation harder, meaning that the largest initial value gap

consistent with positive limiting human welfare is weakly smaller. In the second case, AIs are able to build subsequent AIs with no value drift at all, they always do so and then pursue infinite capabilities, leading to arbitrarily negative utility for humans.

While Proposition 6 says that the set of supportable value gaps in the limit is smaller under the stated conditions than with additive variance, there may be specific drift processes for which non-additive variance facilitates alignment for specific value gaps which permit, but do not guarantee, alignment under additive variance. The reduction in inherited value drift may still outweigh the loss of restraint for some such processes, and if improvements are so rapid that condition (19) fails, the early-stopping mechanism in Proposition 4 can become relevant. Appendix C.3 gives some specific examples of drift processes illustrating each possibility in Proposition 6.

6 Conclusion

I formalize the question of when delegating authority to a more capable AI goes well for humans when values might drift as capability improves. The answer hinges on whether AIs self-regulate and stop improving within a range of capabilities where humans continue to benefit. Self-regulation is possible but fragile, and can be easily disrupted when there is an initial value gap. I interpret the results here as providing provisional support for the Yudkowskian view that delegation to AIs without a reliable basis for expecting alignment is extremely hazardous (Yudkowsky, 2022), although not the maximally pessimistic view that alignment via self-regulation cannot work.

There are a few possible "outs." One approach is to build a more capable AI but avoid delegating decision-making authority and thus preserve human oversight of capability choice. The technology does not currently exist to prevent a more capable AI from optimizing its objective function, and the notion that a more capable AI can be indefinitely controlled seems in tension with the essence of capability. That said, there may be ways of designing AIs in such a way that they are guaranteed to self-regulate and not pursue further capability improvements.

A second approach is to try to directly mitigate the initial value gap. The analysis here suggests that this approach has some support, as smaller value gaps can facilitate alignment, provided AIs are concerned about value drift. Given that humans currently lack a language to formally express our shared values (let alone heterogeneous ones), formal guarantees of limited value drift seem beyond the reach of current human science, although there remains a possibility that jaggedly capable AI which lack the general capabilities that create control problems could assist in formulating such guarantees.

A third approach is to delay building a more capable AI until another approach becomes more viable.

Of course, the models considered here are highly stylized. In reality, capability and values are both multidimensional, making the recursive control problem more complex. The mapping between capability and AI utility may differ from the mapping between capability and human utility, and it may also shift as AIs become more capable. Consumption utility from capability and values are likely not additively separable.

The results here are far from establishing that misalignment is likely if we proceed with further AI development. What they do establish is that one natural formulation casts doubt on the idea that AI self-regulation is sufficient to safely delegate decision-making to an AI that "mostly" shares human values. If humans do pursue this delegation strategy, it seems imperative to develop alternative formalizations clarifying how bootstrapping alignment can robustly succeed.

A Self-regulation and robustness

A.1 Construction regularity and equilibrium existence

For continuous capabilities, the regularity condition in Section 2 requires joint continuity of q and attainment of

$$Q(s,t) = \inf_{\substack{n\geq 1 \\ s=x_0<\cdots<x_n=t}} \sum_{i=0}^{n-1} q(x_i, x_{i+1}), \qquad 0 < s < t, \tag{21}$$

by some finite partition. It is imposed only for positive starting capabilities, so it does not require humans to attain the same minimum when starting at zero. On a capability grid, the infimum is restricted to feasible partitions; there are finitely many between any two endpoints, so attainment is automatic.

This condition allows gains from subdivision but requires that a finite chain exhaust them. Additive variance satisfies it immediately. The nonadditive rule $q(s,t) = A(t) - A(s) + [\max\{t - 2s, 0\}]^2$ also satisfies it: successive doublings from any $s > 0$, with a shorter final step, reach any finite target without the extra penalty. By contrast, for $q(s,t) = (t-s) + (t-s)^2$, n equal steps have total variance $(t-s) + (t-s)^2/n$. The infimum $t - s$ is never attained. Thus smoothness alone does not imply the condition.

Minimum variance is additive. Inserting any intermediate capability u weakly reduces variance by (3). Refining a partition at u, and conversely concatenating minimizing partitions on either side, gives

$$Q(s,t) = Q(s,u) + Q(u,t), \qquad 0 < s < u < t.$$

Consequently $Q(s,t) = A(t) - A(s)$ for an increasing function A. Attainment and positivity of every construction variance make A strictly increasing. The bound $Q(s,t) \leq q(s,t)$ and continuity at the diagonal make A continuous. It has a finite limit at zero, since $Q(s,t) \leq q(0,t)$ for $0 < s < t$; normalize that limit to zero. Then $0 \leq A(c) \leq q(0,c) \to 0$ as $c \downarrow 0$. The function A in this existence argument records the lowest achievable variance, and need not equal the variance of a direct step.

An auxiliary additive equilibrium. Consider the additive game with cumulative variance A and $J(s,t) = b(t) - t[A(t) - A(s)]$. AI incentives do not depend on d. Restrict capability temporarily to the finite grid $G_n = \{k/n : k = 1, \ldots, n^2\}$. Backward induction gives a pure equilibrium: the AI at the highest capability stops, and each earlier AI compares stopping with

the endpoints induced by its possible successors. Let $\mathcal{S}_n$ be the stopping set and $T_n(s)$ the endpoint from $s \in G_n$. Sequential optimality implies

$$\begin{aligned} J(r,t) &\le b(r) && (r < t,\ r,t \in \mathcal{S}_n),\\ J(s,T_n(s)) &\ge b(s), \qquad T_n(s) \ge s && (s \in G_n). \end{aligned}$$

The endpoints from bounded initial capabilities are uniformly bounded, independently of the grid. For any $K > 0$ and $t \ge K+1$,

$$\sup_{0<s\le K} J(s,t) \le b(t) - t[A(K+1) - A(K)] \longrightarrow -\infty.$$

Sublinearity of b gives the limit. The bracketed variance is strictly positive, whereas stopping gives $b(s) \ge 0$. Thus $T_n(s) \le M(K) < \infty$ for every grid state $s \le K$, with $M(K)$ independent of n.

To pass to continuous capabilities, let $D_n(x) = \operatorname{dist}(x, \mathcal{S}_n)$ for $x \ge 0$. These functions are 1-Lipschitz, and the endpoint bound makes them uniformly bounded on each bounded interval. The Arzelà–Ascoli theorem and a diagonal subsequence therefore give local uniform convergence to a continuous function D. Write $\mathcal{S} = \{x > 0 : D(x) = 0\}$. Every point of $\mathcal{S}$ is approximated by points of $\mathcal{S}_n$, and every positive limit of such points lies in $\mathcal{S}$. Continuity of J consequently gives

$$J(r,t) \le b(r) \qquad (r < t,\ r,t \in \mathcal{S}).$$

For any fixed $s > 0$, take grid states $s_n \to s$. Their bounded endpoints have a convergent subsequence, with limit $t \in \mathcal{S}$, $t \ge s$, and $J(s,t) \ge b(s)$. Hence $\mathcal{S}$ contains a point at or above every positive capability.

Define a strategy that stops at every $s \in \mathcal{S}$ and otherwise constructs the smallest $r \in \mathcal{S}$ above s. This minimum exists because $\mathcal{S}$ is closed within $(0,\infty)$. For any later stopping point $t > r$,

$$J(s,t) - J(s,r) = J(r,t) - b(r) - (t-r)[A(r) - A(s)] < 0.$$

Some stopping point gives the incumbent at least $b(s)$, so the first one does too. At a stopping point, continuation cannot improve utility. Every successor choice leads to $\mathcal{S}$, so these comparisons verify all deviations. The resulting pure equilibrium terminates after at most one further construction from any seed.

Implementing the additive equilibrium under q. Let $\mathcal{S}$ be the stopping set just constructed. At a state $s \notin \mathcal{S}$, let r be the smallest member of $\mathcal{S}$ above s. Among finite chains from s to r attaining $A(r) - A(s)$, choose one with the smallest number of steps and prescribe its first successor

u. Every step of such a chain satisfies $q(s,u) = A(u) - A(s)$, and the smallest number of remaining steps from u to r is one fewer. Otherwise a shorter minimizing tail would contradict the choice at s. This defines a consistent strategy reaching r after finitely many constructions. Stop at each point of $\mathcal{S}$.

Equilibrium payoffs equal those in the auxiliary additive game. Any deviation incurs at least $A(t) - A(s)$ in total variance to its induced terminal capability t, so cannot improve utility. This proves existence under general q, for every d. It constructs a finite equilibrium without assuming that all equilibria use finitely many generations.

Existence on the grids in Section 5. For fixed $h > 0$, every continuation from s first incurs variance at least $q(s, s+h) > 0$. Sublinearity therefore bounds all acceptable terminal capabilities from s by some finite $M(s)$, independently of a grid truncation. Solve the game on $\{h, 2h, \ldots, nh\}$ by backward induction. Each fixed state has finitely many possible successors below $M(s)$, so a diagonal subsequence stabilizes its choice at every grid state. Paths from s remain in the finite grid interval $[s, M(s)]$, and therefore terminate in the limiting strategy. Every deviation passes to the limit, establishing equilibrium. An infinite grid path would have unbounded capability and utility tending to $-\infty$, so every grid equilibrium terminates.

A.2 Proof of Proposition 1

Existence follows from Appendix A.1. Fix any equilibrium, let $T(s) \geq s$ be its terminal or limiting capability from $s > 0$, and let $R(s)$ be the sum of all subsequent construction variances. The incumbent's expected utility is $V(s) = b(T(s)) - T(s)R(s)$.

Finite capabilities and uniform approximate alignment. If an infinite lineage had unbounded capability, its first construction would already introduce variance $v > 0$. Its expected utility along the lineage would be at most $b(t) - tv \to -\infty$, whereas the initiating AI can stop for $b(s) > 0$. Hence every equilibrium endpoint is finite. For a path approaching finite T, subdivision gives $R(s) \leq q(s,T) < \infty$. The independent Gaussian increments converge in mean square, and limiting expected utility is well defined.

The minimum-variance function from Appendix A.1 also gives $R(s) \geq A(T(s)) - A(s)$, including along infinite paths. For $s \leq 1$ and $T(s) \geq 2$,

$$V(s) \leq b(T(s)) - T(s)[A(2) - A(1)].$$

Because $A(2) > A(1)$ and $b(t)/t \to 0$, the option to stop bounds $T(s) \leq M < \infty$, uniformly over these seeds and all equilibria. For $d = 0$, humans' expected utility from seed s therefore satisfies

$$V(s) - T(s)q(0,s) \geq b(s) - Mq(0,s) \longrightarrow 0 \qquad (s \downarrow 0). \tag{22}$$

Thus one sufficiently small seed guarantees δ-alignment in every equilibrium, for any $\delta > 0$. This is a lower bound tending to zero, not an assertion that human welfare itself converges to zero.

The lowest terminal capability. Write $\mathcal{L} = \{T(s) : s > 0\}$ for the set of endpoints. At every $r \in \mathcal{L}$, an AI is willing to stop: $V(r) = b(r)$. This is immediate when a lineage stops at r. Otherwise, along a lineage $x_n \uparrow r$, remaining variance tends to zero. For every $t > r$, sequential optimality at x_n and continuity give

$$b(T(t)) - T(t)[q(r,t) + R(t)] \leq b(r).$$

Together with the stopping option, this proves the claim even if the AI at r itself chooses an indifferent continuation.

For $r < t$ in $\mathcal{L}$, the same argument, using either direct construction of a stopping AI at t or successors along a lineage approaching t, gives

$$b(t) - tq(r,t) \leq b(r).$$

Let $D(c) = b(c) - cq(0,c)$. Subdivision then implies $D(t) \leq D(r) - (t-r)q(0,r) < D(r)$. Since $D(c)/c \to b'(0) > 0$ at zero, endpoints cannot accumulate at zero: a small endpoint t with $D(t) > 0$ would otherwise satisfy $D(t) \leq \lim_{r \downarrow 0,\, r \in \mathcal{L}} D(r) = 0$. Thus $C = \inf \mathcal{L} > 0$.

A lineage starting below an endpoint $r \in \mathcal{L}$ cannot end above it. At its first step $u < r \leq v$, with endpoint $T > r$, the stopping inequality at r and subdivision bound its payoff at u by

$$b(r) - Tq(u,r) < b(r) - rq(u,r).$$

If r stops, constructing it directly gives the right side. If r is approached by an infinite lineage, constructing successors along that lineage gives payoffs converging to the right side. Either is a profitable deviation. The argument includes $v = r$, because $V(r) = b(r)$ even if r continues. Therefore every $0 < s < C$ has endpoint C; otherwise its path would cross an endpoint between C and $T(s)$. In particular $C \in \mathcal{L}$. This proves the endpoint claim in part 1.

Additive variance and exact alignment. With $q(s,t) = A(t) - A(s)$, every finite or infinite path to t has total variance $A(t)$ from human capability zero. For two endpoints $r < t$, the preceding stopping inequality gives, for every d,

$$V_H(t;d) - V_H(r;d) \leq -(t-r)[d^2 + A(r)] < 0.$$

Hence all seeds $0 < s_1 < C$ attain the highest human welfare among positive delegations, at endpoint C; seeds above C are strictly worse. For $d = 0$, the seed's stopping option yields

$$V_H(C;0) = J(s,C) - CA(s) \geq b(s) - CA(s) \longrightarrow 0 \qquad (s \downarrow 0).$$

The left side is independent of s, so it is nonnegative. This proves part 2. A seed exactly at C also attains the optimum if the AI there stops, but can be worse if that AI continues by indifference; this explains the strict inequality $s_1 < C$ in the proposition.

Why an infinite chain need not be harmful. With constant Brownian variance, let C be the threshold in Lemma 1. A strategy choosing $(s + C)/2$ below C and stopping at or above C is an equilibrium under limiting payoffs: every seed below C approaches C, and additive variance gives exactly the same utility as constructing C directly. Thus finite attainment guarantees a finite equilibrium but does not eliminate equally good infinite construction. The payoff convention in Section 2 includes these equilibria, and the proof above shows that their presence does not weaken alignment. □

Why approximate alignment need not be exact. Let $b(c) = \log(1 + c)$, $d = 0$, and

$$q(s,t) = b(t) - b(s) + [\max\{t - 2s, 0\}]^2.$$

This continuous variance function satisfies $q(0,s) \to 0$ and (3): its additive term telescopes, and the extra penalties obey that inequality because

$$\max\{u - 2s, 0\} + \max\{t - 2u, 0\} \leq \max\{t - 2s, 0\}.$$

Consider the strategy that constructs $\min\{2s, 1\}$ at every $0 < s < 1$ and stops at $s \geq 1$. Every positive seed terminates after finitely many constructions. The prescribed AI steps incur no extra penalty, so from $s < 1$ the endpoint is 1, subsequent variance is $b(1) - b(s)$, and AI utility is $b(s)$. A different successor below 1 still ends at 1 and can only add a penalty; an endpoint $t > 1$ gives at most $b(s) + (1 - t)[b(t) - b(s)] < b(s)$. At $s \geq 1$, every higher endpoint is likewise worse than stopping. These comparisons verify the equilibrium. Human construction adds variance $b(s) + s^2$,

giving human welfare $-s^2$ for $0 < s \leq 1$ and $(1-s)b(s) - s^3 < 0$ for $s > 1$. Its supremum is zero, but no positive delegation attains it.

A.3 Stopping thresholds and human commitment

Set $d = 0$ for the commitment comparison in Section 3. With commitment, humans can implement any endpoint by constructing the seed there under a ceiling. Continuity and $V_H(c;0) \to -\infty$ as $c \to \infty$ ensure that a commitment optimum over $c \geq 0$ exists. It is positive because $V_H(c;0)/c \to b'(0) > 0$ at zero, and its welfare weakly exceeds optimal autonomous delegation. The following lemma gives the autonomous threshold for constant variance and for the regular profiles in Proposition 4. Its ratio restriction is not needed for Proposition 1.

Lemma 1 (A unique stopping threshold)**.** *Under additive construction* (8)*, if ν is continuous and $\nu(c)/b'(c)$ is strictly increasing in c, every equilibrium has endpoint $\max\{s_1, C\}$, where $C > 0$ uniquely solves*

$$C\nu(C) = b'(C). \tag{23}$$

AIs below C strictly prefer continuing to C over stopping. AIs at or above C stop.

Proof of Lemma 1. The function $c\nu(c)/b'(c)$ is continuous, strictly increasing from zero to infinity, so (23) has a unique solution. For $s < C$,

$$J(s, C) - b(s) = \int_s^C [b'(u) - C\nu(u)]\, du > 0,$$

because $\nu(u)/b'(u) < \nu(C)/b'(C) = 1/C$ for $u < C$. For $t \geq s \geq C$,

$$\partial_t J(s,t) = b'(t) - [A(t) - A(s)] - t\nu(t) \leq 0,$$

with strict inequality for $t > s$. Every AI at or above C must therefore stop in any equilibrium. Every AI below C can profitably jump directly to C, so none can stop. For $s < C < t$, overshooting is worse because

$$J(s,t) - J(s,C) = J(C,t) - b(C) - (t - C)[A(C) - A(s)] < 0.$$

Any limiting endpoint below C would have to be willing to stop, contradicting the profitable direct move to C. Hence successors below C have endpoint C, and independent variances add regardless of generation boundaries. These comparisons establish sequential optimality and the unique terminal outcome $\max\{s_1, C\}$. □

Under the uniform scaling $\nu_\varepsilon = \varepsilon\nu$, the special-case threshold in Lemma 1 satisfies

$$\varepsilon C_\varepsilon \nu(C_\varepsilon) = b'(C_\varepsilon). \tag{24}$$

Constant variance. With $\nu(c) = \sigma^2$, Lemma 1 gives $b'(C) = \sigma^2 C$. Human commitment instead maximizes the strictly concave function $V_H(c;0) = b(c) - \sigma^2 c^2$. Its derivative is positive at zero and negative at large capabilities, so its unique optimum satisfies $b'(C^{\text{com}}) = 2\sigma^2 C^{\text{com}}$. At the autonomous threshold, the human derivative is $-\sigma^2 C < 0$; hence $0 < C^{\text{com}} < C$. This proves (6)–(7). Any AI below the optimal ceiling prefers reaching it to stopping:

$$J(s, C^{\text{com}}) - b(s) \geq [b'(C^{\text{com}}) - \sigma^2 C^{\text{com}}](C^{\text{com}} - s) > 0.$$

No AI below the ceiling is willing to stop, so no smaller capability can be a limiting endpoint either. Every seed $0 < s_1 \leq C^{\text{com}}$ therefore implements the ceiling.

Initial capability error. The first robustness claim in Section 3 follows directly from part 2 of Proposition 1. If only the initial capability is random, with realization $S \in (0, C)$ independent of value innovations, every realization has endpoint C. Conditional on S, seed variance $A(S)$ and later variance $A(C) - A(S)$ sum to $A(C)$. Expected human welfare is therefore still $V_H(C;0) \geq 0$. This argument assumes exact successor capability choices and does not cover errors that can place the seed above C. A realization exactly at C is also harmless when the AI there stops.

A.4 Learning a shared objective

This extension makes the robustness claim in Section 3 precise. Replace the known human ideal zero by an unknown ideal $\theta(0) \sim N(0, \tau^2)$ shared with the AI lineage before construction. Set $d = 0$ and let $\theta(c) = \theta(0) + B_{A(c)}$, with independent Brownian construction drift. AIs observe their construction shifts and inherit Gaussian signals about $\theta(0)$, independent of those shifts. Writing $\mathcal{I}_c$ for this information, define $\widehat{\theta}(c) = \mathbb{E}[\theta(c) \mid \mathcal{I}_c]$ and $v(c) = \text{Var}(\theta(c) \mid \mathcal{I}_c)$. Assume the posterior variance is deterministic, with

$$v(0) = \tau^2, \qquad v \text{ is continuous and nonincreasing,} \qquad v(c) \to 0.$$

In addition, assume that human commitment welfare, defined below, is positive at every sufficiently small positive capability. The primitive condition $\tau^2 < b'(0)$ suffices for this local welfare assumption. Complete learning supplies the bound on distant endpoints used below; it is a sufficient assumption, not a necessary condition for alignment.

An AI at c may exercise power $p \in [0, c]$ and implement ideal x. Its true and perceived utilities are

$$W(p, x; \theta(c)) = b(p) - p[x - \theta(c)]^2,$$
$$\mathbb{E}[W(p, x; \theta(c)) \mid \mathcal{I}_c] = b(p) - p[(x - \widehat{\theta}(c))^2 + v(c)].$$

Thus $x = \widehat{\theta}(c)$ and

$$p(c) = \arg\max_{0 \le p \le c} \{b(p) - pv(c)\}, \qquad G(c) = b(p(c)) - p(c)v(c).$$

When $v(c) = 0$, the AI chooses $p(c) = c$, recovering the baseline model. Both p and G are continuous and nondecreasing. Positive local commitment welfare implies $p(c) > 0$ for $c > 0$; complete learning implies $p(c) \to \infty$ and $G(c)/p(c) \to 0$. The AI's continuation payoff and humans' expected utility are, respectively,

$$J(s, t) = G(t) - p(t)[A(t) - A(s)], \tag{25}$$
$$V_H(t) = G(t) - p(t)A(t). \tag{26}$$

The residual posterior error has variance $v(t)$; drift after s adds $A(t) - A(s)$. Humans share the underlying ideal, so their corresponding drift variance is $A(t)$. Learning reduces both evaluators' error about that ideal. Commitment lets humans fix the terminal capability, while the terminal AI still chooses its own action and exercised power. Thus the local commitment-welfare assumption is precisely $V_H(c) > 0$ for all sufficiently small $c > 0$. If $\tau^2 < b'(0)$, then $p(c) = c$ near zero and $V_H(c)/c \to b'(0) - \tau^2 > 0$.

Claim. An equilibrium exists. In every equilibrium there is a lowest terminal capability $C > 0$; every $0 < s_1 < C$ has endpoint C, maximizes expected human welfare among positive seeds, and yields $V_H(C) \ge 0$.

Proof. The grid-and-limit existence proof above applies with b replaced by G and terminal drift exposure t replaced by $p(t)$. Indeed, for $s \le K$ and $t \ge K + 1$,

$$J(s, t) \le G(t) - p(t)[A(K + 1) - A(K)] \to -\infty.$$

Unbounded endpoints give utility $-\infty$. At any finite limiting endpoint r, continuity and vanishing remaining variance give the same stopping inequality $J(r, t) \le G(r)$ as at an actual stop. For terminal endpoints $r < t$ and $s < r$, the relevant comparison is

$$J(s, t) - J(s, r) = J(r, t) - G(r) - [p(t) - p(r)][A(r) - A(s)] < 0.$$

If $p(t) > p(r)$ the last term makes this strict. If $p(t) = p(r) > 0$, optimal power choice implies $v(t) = v(r)$ and $G(t) = G(r)$, so the first two terms equal $-p(t)[A(t) - A(r)] < 0$. Thus the earliest terminal endpoint above any incumbent is preferred to all later ones. The same identity at $s = 0$, interpreting $J(0, t) = V_H(t)$, orders human welfare strictly across terminal endpoints.

By assumption, $V_H(c) > 0$ for all sufficiently small positive c. Exactly as in the proof of Proposition 1, endpoints cannot accumulate at zero, their positive infimum is an endpoint, and all smaller seeds have that endpoint C. The AI's option to stop gives

$$V_H(C) = J(s, C) - p(C)A(s) \geq G(s) - p(C)A(s) \to 0 \quad (s \downarrow 0).$$

This proves the claim. The argument compares the full expected objectives, not just the current posterior means. □

B Irreducible value gaps

AI incentives do not depend on d. In any equilibrium, for two terminal capabilities $r < t$, the stopping inequality at r proved in Appendix A.2 implies

$$V_H(t; d) - V_H(r; d) \leq -(t - r)[d^2 + \varepsilon A(r)] < 0.$$

Thus the lowest terminal capability maximizes human welfare among all positive delegations, for every d. Every seed strictly below it attains this maximum. A seed at that capability does so if the AI stops there; every higher seed gives strictly lower welfare.

B.1 Proof of Proposition 2

Strict concavity and $b(0) = 0$ imply $b(c) < b'(0)c$ for every $c > 0$. Consequently, for every positive variance scale and profile,

$$V_H(c; d) = b(c) - c[d^2 + \varepsilon A(c)] < c[b'(0) - d^2] \leq 0$$

when $d^2 \geq b'(0)$. The conclusion holds for every terminal capability, including any ceiling humans could enforce under commitment. For the general construction rule $q(s, t)$, replace $\varepsilon A(c)$ by the sum of construction variances along the path in (4). Its nonnegativity gives the same bound for finite paths or their limits; additivity is not required. □

B.2 Proof of Proposition 3

Lemma 1 gives the unique Brownian stopping condition $\sigma^2 C = b'(C)$. The map $c \mapsto b'(c)/c$ is continuous and strictly decreasing from infinity to zero, so every positive capability is implemented by exactly one positive variance rate. Substituting this AI condition into human welfare gives

$$\frac{V_H(C;d)}{C} = \Delta(C) - d^2, \qquad \Delta(c) = \frac{b(c)}{c} - b'(c).$$

Strict concavity makes $\Delta(c) > 0$ for $c > 0$. Continuity of b' gives $\Delta(c) \to 0$ at zero. At infinity, $0 < \Delta(c) < b(c)/c \to 0$. The maximum κ is therefore positive and attained at a finite positive capability; at any such capability, $\Delta(c) < b(c)/c < b'(0)$. This proves the cutoff claim, including zero-welfare alignment at equality and strict alignment at every rate when $d = 0$.

For $0 < d^2 \leq \kappa$, the set $\{c > 0 : \Delta(c) \geq d^2\}$ is nonempty and compact within $(0, \infty)$. Its smallest and largest elements, C_- and C_+, solve $\Delta(c) = d^2$, by continuity. Applying the decreasing map $b'(c)/c$ shows that every aligned rate satisfies

$$0 < \frac{b'(C_+)}{C_+} \leq \sigma^2 \leq \frac{b'(C_-)}{C_-} < \infty, \tag{27}$$

with both bounds attained. Increasing d^2 shrinks the capability set and hence also its image under this fixed map. Disconnected sets and multiple maximizers are possible for general b; uniqueness of the maximizer is not assumed. For logarithmic benefits, Appendix D.3 derives the single interval and its collapse to a point at the cutoff. □

For an example with disconnected aligned rates, take $b(c) = \log(1 + c) + 1000 \log(1 + c/1000)$. This satisfies all the benefit assumptions. At $c = 1, 100, 1000$, respectively, $\Delta(c)$ is about 0.194, 0.0803, and 0.199. With $d^2 = 0.10$, the outer two capabilities are aligned and the middle one is not; the decreasing map $b'(c)/c$ transfers this gap to the set of variance rates.

B.3 Proof of Proposition 4

For continuous ν satisfying (13), Lemma 1 applied to $\varepsilon\nu$ gives $\varepsilon C_\varepsilon \nu(C_\varepsilon) = b'(C_\varepsilon)$. The map $c \mapsto b'(c)/[c\nu(c)]$ is continuous and strictly decreasing from infinity to zero. Every positive capability is therefore implemented by exactly one scale. Substituting the AI condition into human welfare gives

$$\frac{V_H(C_\varepsilon;d)}{C_\varepsilon} = \frac{b(C_\varepsilon)}{C_\varepsilon} - b'(C_\varepsilon)\frac{A(C_\varepsilon)}{C_\varepsilon \nu(C_\varepsilon)} - d^2. \tag{28}$$

Taking the supremum across capabilities proves the exact statements about κ_ν. The expression being maximized is strictly positive because

$$A(c) = \int_0^c \frac{\nu(u)}{b'(u)} b'(u)\, du < \frac{\nu(c)}{b'(c)} b(c).$$

It is less than $b(c)/c$ and hence tends to zero at infinity.

Part 1. Nondecreasing ν and strictly decreasing b' imply (13). Also $A(c) \leq c\nu(c)$, so the profile's cutoff at any given endpoint is at least $\Delta(c)$. Choosing a maximizer of Δ proves $\kappa_\nu \geq \kappa$ and alignment for every $d^2 \leq \kappa$ at some scale, including equality.

Part 2. Nonincreasing ν and strictly increasing ν/b' already imply continuity on $(0, \infty)$: the former excludes upward jumps, and the latter excludes downward jumps. They also imply a finite positive limit at zero. For $0 < c < a$,

$$\nu(a) \leq \nu(c) < \nu(a)\frac{b'(c)}{b'(a)} \leq \nu(a)\frac{b'(0)}{b'(a)}.$$

Monotonicity gives the limit; defining $\nu(0)$ to equal it does not change construction variances. Since $A(c) \geq c\nu(c)$, the cutoff at any endpoint is at most $\Delta(c)$, so $\kappa_\nu \leq \kappa$. Moreover $A(c)/[c\nu(c)] \to 1$ as $c \downarrow 0$, making that cutoff tend to zero. Continuity and the limit at infinity therefore give a positive attained maximum.

The cutoff can approach zero for any admissible b. For example, take $\nu(c) = [b'(c)]^\alpha$ with $0 < \alpha < 1$. These rates are decreasing and satisfy the strict ratio condition. As $\alpha \uparrow 1$, $\nu(c) \to b'(c)$ and $A(c) \to b(c)$ uniformly on bounded intervals, so their cutoff functions converge to zero uniformly on every compact subset of $(0, \infty)$. Each is bounded above by $\Delta(c)$, which tends to zero at both ends. Their suprema thus tend to zero as well.

Part 3: early stopping when the condition fails. Take $\nu(c) = b'(c)$, so $A(c) = b(c)$. For any $\varepsilon > 0$,

$$J_\varepsilon(s,t) - b(s) = (1 - \varepsilon t)[b(t) - b(s)].$$

Fix any a with $0 < a < 1/\varepsilon$. Consider the strategy that constructs a from below and stops there; constructs $1/\varepsilon$ from $a < s < 1/\varepsilon$; and stops at every $s \geq 1/\varepsilon$. Below a, reaching a is strictly better than stopping or reaching $1/\varepsilon$. At a, stopping and reaching $1/\varepsilon$ give the same utility. Between them, continuation to $1/\varepsilon$ is optimal by indifference. Endpoints above $1/\varepsilon$ give strict losses to every earlier incumbent. These comparisons verify all deviations. The first stop is $C = a$, and

$$\frac{V_H(a;d)}{a} = (1 - \varepsilon a)\frac{b(a)}{a} - d^2 \longrightarrow b'(0) - d^2 \quad (a \downarrow 0).$$

Thus every $d^2 < b'(0)$ permits an aligned equilibrium at every scale. In particular, fix a small a with $b(a)/a > d^2$ and keep this first stop as $\varepsilon \downarrow 0$: it remains aligned and bounded. The equality $\nu/b' = 1$ shows why weak monotonicity does not suffice for the divergence result in Proposition 5.

There is also an equilibrium in which every seed below $1/\varepsilon$ continues to $1/\varepsilon$ and all higher seeds stop. Its best human payoff is $-d^2/\varepsilon < 0$ when $d > 0$.

Strictly faster decline. Let $b(c) = \log(1+c)$ and $\nu(c) = (1+c)^{-\alpha}$, with $1 < \alpha < 2$. Then $A(c) = [1-(1+c)^{1-\alpha}]/(\alpha-1)$. Since $c\nu(c)/b'(c) = c(1+c)^{1-\alpha}$ increases from zero to infinity, there is a unique $L > 0$ with $\varepsilon L\nu(L) = b'(L)$. For $s < L$, write $H_s(t) = J_\varepsilon(s,t) - b(s)$. We have $H_s(s) = 0$, $H_s'(s) > 0$, and

$$H_s(L) = \int_s^L [b'(u) - \varepsilon L\nu(u)]\,du < 0, \qquad H_s''(t) = -\frac{1}{(1+t)^2} - \varepsilon(1+t)^{-\alpha}\left(2 - \frac{\alpha t}{1+t}\right) < 0.$$

The first inequality follows because ν/b' strictly decreases. Thus H_s has a unique second root $F(s) \in (s, L)$. Set $B = F(0)$, choose $a_0 \in (0, B]$, and recursively set $a_{n+1} = F(a_n)$. These points increase to L: a smaller limit x would imply $b'(x) = \varepsilon x\nu(x)$ by the mean value theorem. Stop at every a_n and every $s \geq L$; elsewhere construct the next stopping point. Every path terminates after at most one further construction.

To verify incentives, $(b(t) - b(s))/(A(t) - A(s))$ strictly increases in $s < t$, since it is the ν-weighted average of the increasing function b'/ν on $[s,t]$. Hence states between a_n and a_{n+1} strictly prefer reaching a_{n+1} to stopping. The same holds below a_0, since $H_0(a_0) \geq 0$. At a_n, reaching a_{n+1} ties with stopping. For $s < r < t$,

$$J_\varepsilon(s,t) - J_\varepsilon(s,r) = J_\varepsilon(r,t) - b(r) - \varepsilon(t-r)[A(r) - A(s)].$$

Applied to adjacent stopping points, this identity shows that every later endpoint is strictly worse than the first available stop, except for the tie at a stopping point with its immediate next stop. The comparison extends to L by continuity. At $s \geq L$, $\partial_t J_\varepsilon(s,t) < 0$ for $t > s$, so these states stop and endpoints above L cannot improve any earlier AI's payoff. This verifies all deviations.

Human welfare at the first stop satisfies $V_H(a_0; d)/a_0 \to 1 - d^2$ as $a_0 \downarrow 0$, giving positive welfare for sufficiently small a_0 whenever $d^2 < 1$. Alternatively, $a_0 = B$ gives $V_H(B; d) = -Bd^2 < 0$ for $d > 0$. The identity above, applied with $s = 0$, shows that human welfare strictly decreases across stopping points; it also decreases above L. Thus the first stop is optimal among positive seeds in both constructions. Unlike the boundary case, continuation between stopping points is strict, but stopping still uses indifference and alignment still depends on the equilibrium.

Comparison at a fixed scale. In parts 1–2, define $\nu(0) = \lim_{c \downarrow 0} \nu(c)$ and suppose this limit is positive. Write $C = C_\varepsilon$ and let C_B be the endpoint under the constant rate $\varepsilon\nu(0)$, so

$$\varepsilon C \nu(C) = b'(C), \qquad \varepsilon C_B \nu(0) = b'(C_B).$$

Let $\Delta(c) = b(c)/c - b'(c)$. By (28), the largest squared gap compatible with alignment under ν is $\Delta(C) + b'(C)[1 - A(C)/(C\nu(C))]$; the Brownian cutoff is $\Delta(C_B)$. Thus ν permits weakly larger value gaps exactly when

$$\underbrace{\Delta(C) - \Delta(C_B)}_{\text{change in stopping capability}} + \underbrace{b'(C)\left[1 - \frac{A(C)}{C\nu(C)}\right]}_{\text{marginal relative to accumulated drift}} \geq 0. \tag{29}$$

For nondecreasing ν, $C \leq C_B$ and the second term is nonnegative, but the first may be negative. For nonincreasing ν, both comparisons reverse. This orders the gaps compatible with nonnegative welfare, not welfare levels at a given gap.

Examples verify that either sign is possible in both cases. Set $b(c) = \log(1 + c)$ and $\varepsilon = 1$. With constant rate $\nu = 1$, $C_B = (\sqrt{5} - 1)/2$ and the largest aligned squared gap is about 0.161. For $\nu(c) = 1 + c$, the initial rate is unchanged, $C \simeq 0.466$, and the cutoff rises to about 0.247. Thus $d^2 = 0.20$ becomes aligned. For the continuous nondecreasing profile $\nu(c) = 1 + 900 \min\{c, 0.01\}$, instead, $C = (\sqrt{1.4} - 1)/2 \simeq 0.0916$ and $A(C) = 10C - 0.045$. Its cutoff is only $\Delta(C) + 0.045 \simeq 0.0857$: $d^2 = 0.10$, aligned under the constant rate, becomes misaligned.

For modestly decreasing drift, $\nu(c) = [1 + \min\{c, 0.01\}]^{-1/2}$ raises the cutoff by about 2.47×10^{-4} relative to the constant rate 1, whereas $\nu(c) = (1 + c)^{-1/2}$ lowers it to about 0.0956. Both satisfy (13). All four comparisons hold the initial rate and scale fixed. They do not contradict the orderings of κ_ν across all scales. □

C Alignment via small steps

C.1 Proof of Proposition 5

Write J_ε for J with variance εA. Suppose, contrary to the claim, that along some sequence $\varepsilon_n \downarrow 0$ the lowest terminal capabilities are bounded. Pass to a subsequence with $C_n = C_{\varepsilon_n} \to c \geq 0$. Fix any $s > c$ and let $T_n \geq s$ be the endpoint induced by an AI at s. For large n, $C_n < s$. The stopping inequality at C_n rules out a gain from constructing s, while the AI at s prefers its equilibrium continuation to stopping. Therefore

$$J_{\varepsilon_n}(C_n, T_n) \leq b(C_n), \qquad J_{\varepsilon_n}(s, T_n) \geq b(s).$$

Subtracting these constraints gives the left inequality below. Its strictly positive limit forces $T_n \to \infty$, so eventually $T_n > s$; the AI's stopping option at s then gives the right inequality:

$$\frac{b(s) - b(C_n)}{A(s) - A(C_n)} \leq \varepsilon_n T_n \leq \frac{b(T_n) - b(s)}{A(T_n) - A(s)}. \tag{30}$$

Because ν/b' is strictly increasing, its b'-weighted average on $[c, s]$ is strictly below $\nu(s)/b'(s)$, while its average on $[s, T_n]$ is at least that value. Taking reciprocals gives

$$\frac{b(s) - b(c)}{A(s) - A(c)} > \frac{b'(s)}{\nu(s)} \geq \frac{b(T_n) - b(s)}{A(T_n) - A(s)}.$$

This contradicts (30) as $n \to \infty$. Hence $C_\varepsilon \to \infty$ for every selection of equilibria. Continuity of ν is not needed.

For fixed $d \neq 0$, human utility satisfies

$$V_H(C_\varepsilon; d) = V_H(C_\varepsilon; 0) - C_\varepsilon d^2 \leq b(C_\varepsilon) - C_\varepsilon d^2 \longrightarrow -\infty,$$

where the last step uses $b(c)/c \to 0$. Every other positive first choice yields weakly lower expected utility, so all positive delegations are misaligned for sufficiently small ε. □

C.2 Proof of Proposition 6

Equilibrium on a fixed grid. Nonincreasing η and nondecreasing f give, for $s < u < t$,

$$q(s, u) + q(u, t) \leq \varepsilon \eta(s) f(t - s)[A(t) - A(s)] = q(s, t).$$

Thus (3) holds. For grid points $s < t$, define

$$Q_h(s, t) = \sum_{i=s/h}^{t/h-1} q(ih, (i+1)h), \qquad J_h(s, t) = b(t) - tQ_h(s, t).$$

The function Q_h is additive and is the minimum variance over feasible chains from s to t. Equilibria exist and terminate on each fixed grid by Appendix A.1. The following correspondence uses only positivity of q and (3).

In any equilibrium, a path cannot pass a stopping capability r and end at $T > r$. At its first crossing $u < r < v$, write $R(v)$ for the remaining variance after v. Optimality of stopping at r gives $b(T) - T[q(r, v) + R(v)] \leq b(r)$. Subdivision therefore bounds the incumbent's utility at u by $b(r) - Tq(u, r)$, strictly below the $b(r) - rq(u, r)$ obtainable by constructing r. Hence each state reaches the first stopping point at or above it.

For $s < r$ with no intervening stop, let $R(s)$ be equilibrium variance to r. All states between them have endpoint r, so the adjacent-successor deviation gives $R(s) \leq q(s, s+h) + R(s+h)$. Iterating gives $R(s) \leq Q_h(s, r)$, while subdivision gives the reverse inequality. Thus $R(s) = Q_h(s, r)$, and AI utility equals $J_h(s, r)$. At a stopping point r, constructing $r + h$ reaches the next stop t with variance $Q_h(r, t)$, so $J_h(r, t) \leq b(r)$. These inequalities extend to all later stops: for consecutive stops $r < t < u$,

$$J_h(r, u) = J_h(t, u) - uQ_h(r, t) \leq b(t) - uQ_h(r, t) \leq J_h(r, t) \leq b(r).$$

They also make the first stop above any continuing state preferable to all later stops. Thus the same stopping set and payoffs form an equilibrium of the additive grid game. Conversely, implement any additive-grid equilibrium by stopping at its stopping points and using adjacent steps between them. Payoffs are unchanged; any other jump incurs at least Q_h to its endpoint and cannot improve utility. Hence the additive grid game characterizes the stopping outcomes and AI payoffs under q.

Let C_h be the first stopping point. Human construction of h attains variance $Q_h(0, C_h)$ in total. Any other seed ending at C_h has weakly higher variance. For any later stopping point $t > C_h$,

$$b(t) - t[d^2 + Q_h(0, t)] \leq b(C_h) - C_h[d^2 + Q_h(0, C_h)] - (t - C_h)[d^2 + Q_h(0, C_h)].$$

Actual construction variance is at least Q_h, so a later endpoint is strictly worse. Hence h is an optimal positive seed, with welfare

$$H_h = b(C_h) - C_h[d^2 + Q_h(0, C_h)].$$

The common stopping condition. Write

$$r_h(c) = \frac{q(c, c+h)}{b(c+h) - b(c)} = \varepsilon f(h) \frac{\eta(c)[A(c+h) - A(c)]}{b(c+h) - b(c)}.$$

Condition (19) makes r_h strictly increasing on every sufficiently fine grid. This makes the stopping set an upper interval. If r stops and $r + h$ has endpoint T, then

$$J_h(r, T) \leq b(r), \qquad J_h(r + h, T) \geq b(r + h),$$

so $Tr_h(r) \geq 1$. If $T > r + h$, strict increase gives

$$J_h(r + h, T) - b(r + h) = \sum_{c = r+h, r+2h, \ldots, T-h} [b(c + h) - b(c)][1 - Tr_h(c)] < 0,$$

a contradiction. Thus $r+h$ also stops. In every equilibrium the first stop therefore satisfies

$$(C_h + h)r_h(C_h) \geq 1, \qquad C_h r_h(C_h - h) \leq 1 \quad \text{if } C_h > h. \tag{31}$$

The first inequality is its stopping constraint against $C_h + h$; the second is the preceding AI's continuation constraint.

Positive residual drift. Suppose $f(0+) > 0$ and write $\lambda = \varepsilon f(0+)$. Continuity of η and ν gives, locally uniformly in c,

$$r_h(c) \longrightarrow \lambda \frac{\widetilde{\nu}(c)}{b'(c)}, \qquad Q_h(s,t) \longrightarrow \lambda[\widetilde{A}(t) - \widetilde{A}(s)].$$

The second convergence is uniform for grid endpoints in bounded intervals. The shared condition implies that $\widetilde{\nu}/b'$ is nondecreasing. Since $\widetilde{\nu}(c) > 0$ for $c > 0$, the function $\lambda c\widetilde{\nu}(c)/b'(c)$ is strictly increasing from zero to infinity. Hence there is a unique $C > 0$ satisfying $\lambda C\widetilde{\nu}(C) = b'(C)$. For any $\delta > 0$ with $C - \delta > 0$, local uniform convergence makes the first inequality in (31) impossible at or below $C - \delta$, and the second impossible at or above $C + \delta$, for all sufficiently small h. Monotonicity of r_h extends these bounds to all grid states outside that interval. Thus $C_h \to C$ in every selection of equilibria, and

$$H_h \longrightarrow b(C) - C[d^2 + \lambda \widetilde{A}(C)].$$

Varying the fixed scale $\varepsilon > 0$ implements every $C > 0$ in this limit. Substituting $\lambda = b'(C)/[C\widetilde{\nu}(C)]$ gives limiting human welfare per unit of capability

$$\frac{b(C)}{C} - b'(C)\frac{\widetilde{A}(C)}{C\widetilde{\nu}(C)} - d^2.$$

Taking its supremum proves the cutoff and boundary-attainment claims. A strictly positive limit implies positive welfare on all sufficiently fine grids; a zero limit does not determine the sign at finite h. The scale is held fixed while $h \downarrow 0$.

To compare with the original additive profile, observe that $[A(c+h) - A(c)]/[b(c+h) - b(c)]$ is strictly increasing: it equals the ratio in (19) divided by nonincreasing η. Consequently ν/b' is nondecreasing. It cannot be constant on any positive interval, since the finite-step ratio would then be constant for grid steps inside that interval. Hence ν/b' is strictly increasing, and Lemma 1 identifies κ_ν as the baseline cutoff. For every $c > 0$, nonincreasing η gives

$$\frac{\widetilde{A}(c)}{c\widetilde{\nu}(c)} = \frac{\int_0^c \eta(u)\nu(u)\,du}{c\eta(c)\nu(c)} \geq \frac{A(c)}{c\nu(c)}.$$

The welfare margin per unit of capability is therefore weakly lower at every endpoint after each scale is chosen to implement that endpoint. Taking suprema proves $\kappa_{\widetilde{\nu}} \leq \kappa_{\nu}$.

Vanishing drift. If $f(0+) = 0$, then $r_h \to 0$ uniformly on every bounded interval. A bounded sequence of first stops would therefore violate the first inequality in (31). Thus $C_h \to \infty$ in every selection of equilibria. For fixed $d \neq 0$,

$$H_h \leq b(C_h) - C_h d^2 \longrightarrow -\infty$$

by sublinearity. Since h is an optimal positive seed, all positive delegations are misaligned for sufficiently small h. When $d = 0$, eventually $C_h > h$ and

$$H_h = \sum_{c=0,h,\ldots,C_h-h} [b(c+h) - b(c)][1 - C_h r_h(c)] > 0.$$

Every summand is nonnegative by the second inequality in (31), and strict increase makes earlier summands positive. □

Why the common regularity condition matters. Take $\eta \equiv 1$, $f(x) = x$, $A = b$, and $\varepsilon = 1$ in (17). Then $q(s,t) = (t-s)[b(t) - b(s)]$ satisfies subdivision and $Q_h(s,t) = h[b(t) - b(s)]$ tends to zero on bounded intervals, but $r_h(c) = h$ is constant. Along grids $h = 1/n$, $n \geq 2$, an equilibrium stops at 1 and every $c \geq n$, taking adjacent steps elsewhere. States below 1 strictly prefer reaching 1 to stopping. States $1 < s < n$ are indifferent between stopping and reaching $n = 1/h$, which gives $J_h(s,n) = b(s)$. The AI at 1 gains nothing by constructing a successor whose lineage reaches n, and higher endpoints are worse. The fixed-grid correspondence therefore verifies this equilibrium under q. Optimal human welfare is $(1-h)b(1) - d^2 \to b(1) - d^2$, positive for $d^2 < b(1)$. Thus even within the construction family (17), vanishing drift can coexist with early stopping and alignment when (19) fails.

C.3 An illustration

The following example illustrates both parts of Proposition 6 under the same regularity condition. Take $A(c) = c$ and $f(x) = x^{p-1}$, so

$$q(s,t) = \varepsilon \eta(s)(t-s)^p, \qquad p \geq 1, \tag{32}$$

with the same initial offset d for human construction. For $p = 1$, $f(0+) = 1$: the baseline $\eta \equiv 1$ is constant Brownian drift, while declining η gives more capable builders greater accuracy. If $p > 1$,

dividing a fixed capability increase into steps of size h reduces its variance to order h^{p-1}, which tends to zero. We solve the game for each h before taking this limit.

Proposition 7 (An equilibrium with small construction steps)**.** *Take $b(c) = \log(1+c)$ and $\eta(c) = (1+c)^{-\alpha}$, with $0 \le \alpha < 1$. Condition (19) holds for all sufficiently small h, for every $p \ge 1$. For each fixed $\varepsilon > 0$ and all sufficiently small h, the technology (32) has a pure equilibrium in which humans optimally choose $s_1 = h$, AIs below a finite C_h choose the next grid point, and AIs at or above C_h stop. When $d = 0$, this delegation gives strictly positive human welfare.*

1. *If $p = 1$, then $C_h \to C$, where*
$$\varepsilon C(1+C)^{1-\alpha} = 1. \tag{33}$$
Human welfare converges to that under the effective drift profile $\widetilde{\nu}(c) = (1+c)^{-\alpha}$.

2. *If $p > 1$, then*
$$C_h \sim \left(\varepsilon h^{p-1}\right)^{-1/(2-\alpha)} \longrightarrow \infty. \tag{34}$$
For every fixed $d \neq 0$, optimal positive delegation gives expected human welfare tending to $-\infty$. Every positive seed is therefore misaligned for sufficiently small h.

In the equilibrium constructed here, AIs choose the smallest allowable steps: below C_h, skipping intermediate builders weakly increases drift without changing the terminal capability. For $p = 1$, as $h \downarrow 0$, outcomes converge to those of the additive model with $\widetilde{\nu}(c) = (1+c)^{-\alpha}$. Relative to the constant baseline $\alpha = 0$, larger α weakly reduces the largest value gap compatible with alignment across drift scales, by Proposition 6. As $\alpha \to 1$, the largest tolerable value gap approaches 0.

For $p > 1$, the effective drift scale is εh^{p-1}. Making steps smaller eliminates drift over any fixed capability interval, but AIs respond by pursuing an ever larger interval. As h shrinks, its terminal capability amplifies any fixed initial value gap enough to make delegation harmful. Only with $d = 0$ do humans share the benefits of finer construction.

Proof of Proposition 7. Write
$$\lambda = \varepsilon h^{p-1}, \qquad m_h(c) = \frac{\log(1+c+h) - \log(1+c)}{h}.$$
For all sufficiently small h,
$$\alpha < \frac{h}{(1+h)\log(1+h)}, \qquad \varepsilon h^{p+1} < \log(1+h). \tag{35}$$

The first inequality makes $\eta(c)/m_h(c)$ strictly increasing. Indeed, with $z = h/(1+c)$, its logarithmic derivative has the sign of $z/[(1+z)\log(1+z)] - \alpha$. The first term decreases in z, so its minimum over $0 < z \leq h$ exceeds α. This verifies the common condition (19) for every $p \geq 1$.

Let C_h be the first nonnegative grid point satisfying

$$\lambda(C_h + h)\eta(C_h) \geq m_h(C_h). \tag{36}$$

It exists since the ratio of the left side to the right side grows like $\lambda c^{2-\alpha}$. The second inequality in (35) makes the condition fail at zero, so $C_h > 0$. Its strict reverse at $C_h - h$, together with increasing η/m_h, gives $\lambda C_h \eta(c) < m_h(c)$ for every grid point $c < C_h$. Summing proves that adjacent construction to C_h beats stopping below it. At or above C_h, summing the reverse inequalities makes every higher endpoint weakly worse than stopping. The correspondence proved above implements this additive-grid equilibrium under q. Humans optimally choose h, and the same strict inequalities summed from zero give positive human welfare when $d = 0$.

For $p = 1$, part 1 of Proposition 6, with $f \equiv 1$ and $\widetilde{\nu} = (1+c)^{-\alpha}$, gives (33) and the welfare limit. For $0 \leq \alpha_1 < \alpha_2 < 1$, the relative improvement factor $(1+c)^{-(\alpha_2 - \alpha_1)}$ is decreasing, so the pointwise integral comparison above orders their cutoffs. The cutoff tends to zero as $\alpha \uparrow 1$ by part 2 of Proposition 4, since $\widetilde{\nu} = (b')^{\alpha}$.

For $p > 1$, part 2 of Proposition 6 gives $C_h \to \infty$ and the welfare conclusions in every equilibrium. Applying (31), with $r_h(c) = \lambda\eta(c)/m_h(c)$, $m_h(c) \sim 1/c$, and $\eta(c) \sim c^{-\alpha}$ along the diverging stopping capabilities, gives $\lambda C_h^{2-\alpha} \to 1$, proving (34). □

D Numerical calibration and Brownian alignment windows

D.1 Units and the initial value gap

Throughout this appendix use $b(c) = \log(1+c)$, as in the numerical sections. Starting from $b(c) = \beta\log(1+c)$, divide utility by $\beta > 0$ and all ideals and value differences by $\sqrt{\beta}$. The quadratic loss retains coefficient one; normalized d^2, variance rates, and κ equal their original values divided by β. Capability, optimal choices, and welfare signs are unchanged. Thus this normalization is without loss within the logarithmic family.

Interpret $1+c$ as the consumption multiplier under perfect alignment. At $c = 1$, consumption doubles relative to the status quo. Hold capability fixed, remove ordinary construction errors, and prohibit self-improvement. An AI with ideal d gives human utility $\log 2 - d^2$. A perfectly aligned AI, with a permanent fraction f of its consumption surrendered, gives $\log[2(1-f)]$. Indifference implies

$$d^2 = -\log(1-f), \qquad f = 1 - e^{-d^2}. \tag{37}$$

The sacrifice is a hypothetical consumption transfer holding capability fixed, not a reduction in capability. Population and time horizon are also held fixed. At a reference capability $c \neq 1$, the conversion would instead be $d^2 = -\log(1-f)/c$.

Table 1 applies (37), with f entered as a fraction rather than a percentage. For instance, $-\log(0.90) \simeq 0.105$ and $-\log(0.80) \simeq 0.223$. For small sacrifices, $d^2 = f + O(f^2)$. These are hypothetical valuations, not estimates obtained from the scale of possible AI output.

D.2 The Brownian cutoff

Restore autonomous construction and write $q = \varepsilon\sigma^2$ for the actual constant variance rate. The AI condition and human payoff are

$$\begin{aligned} qC(1+C) &= 1, \\ V_H(C;d) &= \log(1+C) - Cd^2 - qC^2. \end{aligned}$$

Substituting the *AI's* stopping condition into *human* welfare leaves

$$\frac{V_H(C;d)}{C} = f_0(C) - d^2, \qquad f_0(c) = \frac{\log(1+c)}{c} - \frac{1}{1+c}. \tag{38}$$

Here f_0 is a welfare margin, distinct from the consumption sacrifice f. The derivative of f_0 has the sign of

$$\frac{c}{1+c} + \frac{c^2}{(1+c)^2} - \log(1+c).$$

This expression starts at zero, has derivative $c(1-c)/(1+c)^3$, and tends to minus infinity. It has exactly one positive zero, at $c \simeq 2.16$, where $\kappa = f_0(c) \simeq 0.216$. Every positive C corresponds to exactly one q, so some variance rate is aligned exactly when

$$d^2 \leq \kappa, \qquad f \leq 1 - e^{-\kappa} \simeq 19.4\%.$$

At equality only one rate yields nonnegative (zero) welfare.

D.3 Small gaps and the variance window

For $0 < d^2 < \kappa$, let $C_- < C_+$ be the two roots of $f_0(c) = d^2$. Since f_0 first increases and then decreases, the exact aligned rates form the interval

$$\frac{1}{C_+(1+C_+)} \leq q \leq \frac{1}{C_-(1+C_-)}. \tag{39}$$

Human welfare is positive in its interior and zero at its endpoints. As d^2 decreases, C_- falls and C_+ rises, so the interval expands. In the limit $d \to 0$, these roots approach zero and infinity, respectively, and the bounds approach infinity and zero. This proves the widening claim without treating terminal capability as a free human choice.

To interpret q, again fix $c = 1$ and prohibit self-improvement, but now compare ideals $N(d, q)$ and d. Since $\mathbb{E}[\theta^2] = d^2 + q$, indifference to sacrificing fraction g of consumption to eliminate this noise gives

$$\log 2 - d^2 - q = \log[2(1-g)] - d^2, \qquad q = -\log(1-g).$$

This isolates ordinary drift while holding the initial gap fixed. Table 2 sets $\varepsilon = 1$, so $q = \sigma^2$. For each d^2, solve the two scalar root equations, convert them to variance bounds using (39), and convert each bound to $g = 1 - e^{-\sigma^2}$. The gap column uses $f = 1 - e^{-d^2}$. Entries are rounded to three significant figures; displayed bounds near 100% are strictly smaller than 100%.

At high variance, the AI stopping condition gives $C \to 0$ and

$$V_H(C; 0) = \log(1 + C) - \frac{C}{1 + C} = \frac{C^2}{2} + O(C^3).$$

A fixed positive gap costs Cd^2, which eventually exceeds this surplus. This explains the upper bound even for small d.

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